

Session Report

Smart Biometric Bridge (SBB) — authenticated punch-ingest path

Field	Detail
Project	SmartPRS by Ametecs — HRM, Payroll & workforce compliance (Laravel 11, PHP 8.3, MySQL)
Session date	16 August 2026
Scope	Add a second, authenticated JSON punch-ingest path alongside the existing ZKTeco/eSSL ADMS emulation
Working copies changed	C:\laragon\www\smartprs2 and C:\laragon\www\smartprs-l3
Files added	13 (12 source + 1 test suite)
Files modified	4
Database changes applied	NONE — migrations written but not run
Git commits made	NONE — all work uncommitted in the working tree
Prepared for	Ejaz Hussain, Ametecs India

Read this first

No database was modified and no commit was made during this session. Everything described here is code on disk, waiting for a decision.

The automated test suite has been **written but not executed** — see Section 6. Running it is the acceptance gate before anything is migrated.

Contents

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1. Executive summary

SmartPRS previously received biometric punches by exactly one route: emulating the ZKTeco/eSSL ADMS iClock protocol. That route has no authentication, no tenant binding, no idempotency, no error reporting, and it silently discards any punch whose device PIN is not yet mapped to an employee.

This session added a **second, parallel route** for Smart Biometric Bridge (SBB) — the on-premise Windows service that collects punches from any vendor's device and forwards them over HTTPS. The new route is authenticated, tenant-bound, idempotent at the database level, and returns a verdict for every single punch.

The existing hardware path is untouched

Not one character of PushController or the /iClock routes changed. Existing customer devices are unaffected. A customer may use either route, or both at once on different devices. Both write into the same attendance_logs table, distinguished by the source column (push vs sbb).

1.1 Outcome at a glance

Area	Before	After (on the new path)
Authentication	None	API key, sha256-hashed, scoped, expirable, revocable
Tenant binding	NULL on auto-registered devices	Always taken from the key; never from the request body
Timestamps	Carbon::parse fallback keeps sender offset	Strict naive-local; offset-bearing values rejected
Duplicate protection	Application-side updateOrCreate, no constraint	Two database unique indexes + insertOrIgnore
Unmapped PIN	Punch discarded permanently	Quarantined, reported, replayed when mapped
Response to sender	Always HTTP 200, no detail	Per-punch verdict: accepted / duplicate / pending / rejected
Logging	None	One log line per batch, one warning per rejection
Late-arrival email	Sent inline, blocks the response	Queued job

2. What existed before

2.1 The single ingest path

```
Device ---> GET/POST /iclock/cdata?SN=...      (routes/web.php:92)
            \-> PushController::cdata
                \-> PushController::importAttlog
                    \-> ETimeOfficeService::import
                        \-> attendance_logs
```

Three further pull-based connectors — ETimeOfficeService, ETimeTrackLiteService and GenericApiService — also converge on ETimeOfficeService::import.

2.2 The attendance_logs table

Created by 2026_06_24_000000_ensure_runtime_feature_tables.php:

Column	Type	Note
id	bigint	primary key
tenant_id	bigint, nullable	indexed
company_id	bigint, nullable	indexed
emp_code	string	indexed
emp_name	string, nullable	
log_date	date	indexed
punch_at	datetime	
direction	string(4)	in / out
source	string	default 'biometric'
created_at / updated_at	timestamps	

There was **no unique constraint of any kind** on this table.

2.3 The five defects this work addresses

Each is a real, current behaviour in the codebase, cited by file and line.

D1 — Cross-customer data leak

PushController::cfgForSn returns tenant_id from the device's biometric_configs row. An unknown serial auto-registers with tenant_id = NULL (PushController.php:214). That NULL flows into ETimeOfficeService::matchEmployee(\$full, \$raw, null), and at ETimeOfficeService.php:321 the tenant filter is applied only when \$tid is truthy:

```
if ($tid && Schema::hasColumn('employees', 'tenant_id')) {
    $q->where('tenant_id', $tid);
}
```

With \$tid = null the filter is skipped entirely, so a punch for EMP001 matches whichever customer's EMP001 the database happens to return first. One customer's attendance can be written against another customer's employee.

D2 — Silent timezone corruption

PushController::parseDate tries four fixed formats and then falls back to Carbon::parse(\$v) (PushController.php:178). Carbon::parse **honours** an offset in the string. ETimeOfficeService::import then stores \$p['punch_at']->format('Y-m-d H:i:s') (ETimeOfficeService.php:420), which renders the time in that offset and discards the offset itself. The digits are taken literally.

Worked example

A punch arriving as 2026-08-16 09:41:00+00:00 on an Indian install is stored as 09:41:00, when the employee actually clocked in at 15:11:00 local. A 5h30m error — silent, unlogged, and straight into payroll.

D3 — Permanent loss of unmapped punches

ETimeOfficeService::import, around line 405:

```
if (! $emp) {  
    $unmatched[$full] = ($unmatched[$full] ?? 0) + 1;  
    continue;    // the punch is gone  
}
```

Only a **counter** survives, in biometric_unmapped. The punch itself is discarded permanently. During every go-live — precisely the window in which device IDs have not yet been mapped — real attendance is destroyed and cannot be recovered.

D4 — No idempotency, and a mechanism that manufactures duplicates

De-duplication relied on updateOrCreate with an application-side match (ETimeOfficeService.php:429) and no database constraint behind it. Two concurrent requests both find no row, and both insert.

This is not theoretical. The command attendance:dedupe (DedupeAttendance.php) exists specifically to repair duplicates produced this way in rev173g, and its own docblock records that duplicates "break both the Attendance Report and payroll".

The duplicate-producing loop is worse than a plain race. ETimeOfficeService::import calls LateArrivalService::notifyTouched **inline**, which ends at Mail::raw (LateArrivalService.php:289). On a slow or unreachable SMTP host the HTTP response is blocked until the sender times out — and an at-least-once sender that times out **retries**, creating exactly the duplicate the retry was meant to avoid.

D5 — Zero observability

The ingest path contains no logging whatsoever; every failure mode is a bare continue. When a customer reports "my punches aren't showing", there is nothing on the server to diagnose it with.

3. What changed

3.1 New files (13)

File	Lines	Purpose
database/migrations/ 2026_08_16_000100_create_api_keys_table.php	45	api_keys table
database/migrations/ 2026_08_16_000200_extend_attendance_logs_for_sbb.php	196	Provenance columns + the two unique indexes
database/migrations/ 2026_08_16_000300_create_attendance_pending_table.php	50	attendance_pending quarantine table
app/Services/ApiKeys.php	78	Key minting, hashing, prefix parsing
app/Services/PunchIngestService.php	390	Ingest, time parsing, quarantine, replay
app/Http/Middleware/ApiKeyAuth.php	156	Key authentication + scope check
app/Http/Controllers/Api/PublicApiController.php	118	ping + ingestPunches
app/Http/Controllers/ApiKeyController.php	267	Settings screens: API Keys, Pending Punches
app/Jobs/NotifyLateArrivals.php	55	Queued late-arrival notification
routes/api.php	30	Stateless /api/v1 routes
resources/views/settings/api-keys.blade.php	136	API Keys screen
resources/views/settings/pending-punches.blade.php	74	Pending Punches screen
tests/Feature/SbbIngestTest.php	481	23 feature tests

3.2 Modified files (4)

File	Change	Risk assessment
bootstrap/app.php	Added api: routing + api-key middleware alias	Additive. Loaded on every request, but no existing route or middleware altered.
routes/web.php	Added 5 routes for the two Settings screens	Additive. No existing route modified.
app/Http/Controllers/ BiometricConfigController. php	mapEmployee now calls replayPending	Guarded by hasTable and wrapped in try/catch. Does nothing until migrated.
resources/views/layouts/ app.blade.php	Two sidebar links	Legacy Blade layout only; the main SPA does not use it.

3.3 Database changes (written, not yet applied)

api_keys — new table

Mirrors SmartEPT's design deliberately, plus `expires_at` which SmartEPT lacks. Stores only the sha256 of the secret; the plaintext is shown once at creation and is never recoverable.

Column	Type
id	bigint PK
tenant_id / company_id	bigint, nullable, indexed
name	string — e.g. "SBB — Chennai Plant"
prefix	string(12), indexed — shown in the UI, e.g. sk_prs_ab12
key_hash	string — sha256 of the full secret
scopes	json, nullable — ['ingest','read']
last_used_at / expires_at	timestamp, nullable
active	boolean, default true
created_at / updated_at	timestamps

attendance_logs — extended

All new columns are nullable, so every existing row stays valid and every existing writer keeps working.

- `device_sn` string(64), nullable, indexed
- `device_user_id` string(64), nullable
- `external_id` string(96), nullable — the sender's idempotency key
- `verify_mode` string(24), nullable
- `direction` widened **string(4) → string(8)** so that unknown fits; a truncated direction is a wrong attendance record

Two unique indexes are added:

- `attlog_tenant_external_unique` (`tenant_id`, `external_id`)
- `attlog_natural_unique` (`tenant_id`, `emp_code`, `punch_at`, `source`)

Migration ordering is load-bearing

Production data already contains duplicates on the natural key. The migration runs `attendance:dedupe` **before** creating `attlog_natural_unique`, with an inline fallback if the command is unavailable.

If the index were created first, the migration would abort on any real install. This was verified empirically — see Section 6.3.

attendance_pending — new quarantine table

Columns: id, tenant_id, company_id, device_sn, device_user_id, punch_at, direction, verify_mode, external_id, source, resolved_at (nullable), timestamps. Unique on (tenant_id, external_id).

3.4 The new API contract

```
GET /api/v1/ping                middleware: api-key
POST /api/v1/attendance/punches middleware: api-key:ingest
```

Both sit under `throttle:300,1`. Authentication is by `X-Api-Key: <key>` or `Authorization: Bearer <key>`. The routes live in `routes/api.php`, deliberately outside the web group, so no session is started and no cookie is issued.

GET /api/v1/ping — 200

```
{
  "ok": true,
  "app": "SmartPRS",
  "version": "2026.6.1",
  "tenant_id": 3,
  "company_name": "Acme Textiles Pvt Ltd",
  "scopes": ["ingest"],
  "timezone": "Asia/Kolkata",
  "server_time": "2026-08-16T09:41:00+05:30"
}
```

`company_name` lets an installer standing in a plant confirm the key points at the right customer. `timezone` and `server_time` let SBB detect a clock mismatch before it starts shipping punches into the wrong hour.

POST /api/v1/attendance/punches — 200, a verdict per punch

```
{
  "ok": true,
  "batch": {
    "received": 10, "accepted": 8,
    "duplicates": 1, "pending": 1, "rejected": 0
  },
  "results": [
    { "external_id": "3f9c...", "status": "accepted" },
    { "external_id": "7b2a...", "status": "duplicate" },
    { "external_id": "c41d...", "status": "pending",
      "reason": "EMPLOYEE_NOT_MAPPED" },
    { "external_id": "9e01...", "status": "rejected",
      "reason": "TIME_FORMAT" }
  ]
}
```

Status meanings for the sender

Status	Meaning	Should SBB retry?
accepted	Stored in <code>attendance_logs</code>	No
duplicate	Already stored; the re-send was a no-op	No
pending	Held in quarantine — device PIN not yet mapped	No — the punch is safe
rejected	Not stored. See reason	Only after fixing the cause

Reject reasons: TIME_FORMAT (malformed or offset-bearing timestamp) and VALIDATION (missing or oversized field, invalid direction).

Error body — 401 / 403 / 422

```
{ "error": { "code": "API_KEY_401", "message": "..."} }
```

Codes in use: API_KEY_401, API_KEY_403, VALIDATION_422.

4. Why each change was made, and its impact

#	Change	Fixes	Impact
1	Tenant taken from the API key and always passed to matchEmployee	D1	Cross-customer matching is structurally impossible on this path. An employee code belonging to another tenant now returns pending, not a wrong match.
2	Strict naive-local time parsing; offsets rejected	D2	A malformed timestamp becomes a visible TIME_FORMAT rejection instead of a silent 5h30m payroll error. No timezone is ever guessed.
3	attendance_pending quarantine + replay on mapping	D3	Go-live no longer destroys attendance. Punches are held, reported to the sender, listed for the admin, and released the moment the ID is mapped.
4	insertOrIgnore against two database unique indexes	D4	Idempotency is a database guarantee rather than a hopeful SELECT. Concurrent retries cannot race.
5	Late-arrival mail moved to a queued job	D4	The response no longer blocks on SMTP, removing the timeout-retry-duplicate loop at its source.
6	Log::info per batch, Log::warning per rejection	D5	"My punches aren't showing" becomes diagnosable from the server log alone.
7	Per-punch verdicts in the response	D4, D5	An at-least-once sender knows exactly what landed. No silent discards.

4.1 Impact on existing behaviour

Unchanged

/iclock endpoints, PushController, ETimeOfficeService::import, ETimeTrackLiteService, GenericApiService, all pull-based syncs, all existing attendance_logs rows, and all reports and payroll logic.

Changed for all writers, after migration

The two unique indexes constrain attendance_logs globally, including the legacy path. This was checked rather than assumed: ETimeOfficeService::import uses updateOrInsert whose match keys are (tenant_id, emp_code, punch_at, source) — exactly attlog_natural_unique. It therefore updates rather than inserts and cannot violate the constraint. Verified empirically in Section 6.3.

One-time destructive step

attendance:dedupe deletes duplicate rows, keeping the newest of each group. This is the existing, already-shipped repair command; the migration invokes it because the unique index cannot be built otherwise.

4.2 Deliberate deviation from the specification

Null-tenant keys are refused on the ingest scope

api_keys.tenant_id is nullable, per SmartEPT parity as specified. **However**, ApiKeyAuth refuses

a null-tenant key on the `ingest` scope with a 401.

Reason: both `attendance_logs` unique indexes include `tenant_id`, and NULL never collides in a unique index in either MySQL or SQLite. A tenant-less key would therefore silently lose its de-duplication guarantee and re-insert every retried punch — the exact defect being fixed. Refusing the key is the only way the idempotency promise is actually true.

Blast radius: none. `InstallController` creates a tenant row even on on-prem installs (`InstallController.php:192`), and the admin UI stamps the creating user's tenant onto every key.

This is flagged for your explicit approval. Say the word and it can be removed.

5. Resulting architecture

PATH 1 – existing, UNCHANGED (no credential possible: firmware speaks it)

```
ZKTeco / eSSL device
  | GET/POST /iclock/cdata?SN=...
  v
PushController --> ETimeOfficeService::import --> attendance_logs
                                                    source = 'push'
```

PATH 2 – new, AUTHENTICATED

```
Any vendor device --> Smart Biometric Bridge (on-prem Windows service)
  | POST /api/v1/attendance/punches X-Api-Key: sk_prs_....
  v
ApiKeyAuth (tenant + company + scopes from the KEY)
  v
PublicApiController::ingestPunches
  v
PunchIngestService::ingest
  |-- parseNaiveLocal ..... offset-bearing -> rejected
  |-- matchEmployee(.., tenantId) ... always tenant-scoped
  |-- matched -> insertOrIgnore -> attendance_logs source = 'sbb'
  |-- unmatched -> insertOrIgnore -> attendance_pending
  `-- NotifyLateArrivals::dispatch .. QUEUED, never inline Mail

Admin maps the device ID (Settings -> Pending Punches)
  v
BiometricConfigController::mapEmployee
  v
PunchIngestService::replayPending -> attendance_logs, resolved_at stamped
```

Both paths write to the same `attendance_logs` table, so attendance and payroll see every punch together regardless of how it arrived. The `source` column distinguishes them.

6. Testing summary

The full test plan, all 23 test cases with steps and expected results, and the sign-off sheet are in the companion document "**SmartPRS — Test Plan & Test Report (SBB Ingest Path)**". This section summarises the position.

6.1 Status

The automated tests have NOT been executed

tests/Feature/SbbIngestTest.php contains 23 test cases and is complete, but it has not been run. Packagist was blocked from the environment the work was done in, so no Laravel harness could be built, and there was no shell access to the Windows machine.

Running ``vendor\bin\pest tests/Feature/SbbIngestTest.php`` is the real acceptance gate. If it fails, do not migrate.

6.2 Coverage written

Area	Cases	Covers
Health check	1	ping returns the right customer, scopes, timezone
Happy path	1	Punch stored with correct employee, tenant, source, device fields
Idempotency	2	Same external_id twice; same moment under a new external_id
Quarantine & replay	2	Unmapped PIN held not lost; mapping releases it
Time handling	3	Offset rejected; 6 further bad formats; wall clock stored literally
Tenant isolation	3	Key A cannot write to tenant B; body tenant_id ignored; foreign employee held
Authentication	7	Missing / unknown / expired / revoked / wrong-scope; Bearer header; last_used_at
Batch behaviour	3	Mixed batch verdicts; individual rejection; empty and oversized batches
Regression	1	/iclock ADMS handshake and ATTLOG upload still work

6.3 What was verified in this session

Four things were checked directly rather than assumed.

(a) Syntax

php -l clean on all 17 PHP files. Blade directives balance: 8 @if / 8 @endif, 2 @foreach / 2 @endforeach, 1 @section / 1 @endsection.

(b) Time-parsing rules — 45 cases, all passing

The parser was extracted and run against 15 inputs across 3 timezones (Asia/Kolkata, UTC, America/New_York). Every case behaved as specified.

Input	Result
2026-08-16 09:41:00	accepted, unchanged
2026-08-16 09:41:00+05:30	rejected — TIME_FORMAT
2026-08-16 09:41:00-08:00	rejected — TIME_FORMAT
2026-08-16T09:41:00Z	rejected — TIME_FORMAT
2026-08-16T09:41:00+0530	rejected — TIME_FORMAT
2026-08-16 09:41:00 UTC / IST	rejected — TIME_FORMAT
16/08/2026 09:41:00	rejected — TIME_FORMAT
2026-02-30 09:41:00	rejected — roll-over guard
2026-13-01 / 25:00:00 / 09:41 / empty	rejected — TIME_FORMAT

(c) Dedupe and unique-index semantics — verified in real SQLite

A table matching the post-migration schema was built and exercised. Actual output:

```
rows before dedupe: 3
index refused while duplicates exist (as designed)  <-- proves the ordering matters
rows after dedupe: 2
both unique indexes created OK
legacy rows with NULL external_id coexist: 4          <-- history unaffected
1st send                -> inserted=1 (accepted)
identical re-send       -> inserted=0 (duplicate)
same moment, new ext id -> inserted=0 (duplicate)
same ext id, other tenant -> inserted=1 (accepted)
legacy updateOrInsert   -> updated=1 (no constraint violation)
EMP1043 in tenant 1: 1
```

The second line is the important one: creating the natural-key index **fails** while duplicates are present. That is the empirical justification for running attendance : dedupe first, and for backing up before you do.

(d) Compatibility checks

- All four modified files were byte-identical between smartprs2 and smartprs - l3 before editing, so the same content is correct for both.
- The code is PHP 8.2-compatible. smartprs - l3 pins PHP 8.2 while smartprs2 requires ^8.3; no 8.3-only syntax was used.
- The new routes app/api-keys and app/pending-punches match no pattern in Edition::BLOCK_SAAS, so EditionGuard will not block them on on-prem builds.

7. Update instructions

7.1 Local / development — C:\laragon\www\smartprs2

```
cd C:\laragon\www\smartprs2

:: 1. See exactly what changed
git status
git diff

:: 2. Run the test suite -- THIS IS THE ACCEPTANCE GATE
vendor\bin\pest tests/Feature/SbbIngestTest.php

:: 3. Only if green, apply the schema
php artisan attendance:dedupe --dry      :: look first; deletes nothing
php artisan attendance:dedupe           :: deletes duplicates
php artisan migrate

:: 4. Clear caches (new routes + new middleware alias)
php artisan optimize:clear

:: 5. Full suite, to confirm nothing else regressed
vendor\bin\pest
```

If the tests fail, **do not migrate**. Send the failure output back for diagnosis.

7.2 Live / production server

Maintenance window required

Step 3 **deletes rows**. Step 4 locks attendance_logs while the indexes build — on a large table that can take several minutes.

Do not run this during a working shift.

```
# 0. BACK UP THE DATABASE FIRST -- non-negotiable
mysqldump -u root -p smartprs > smartprs_backup_2026-08-16.sql

# 1. Stop the biometric feed so nothing writes mid-migration
#   (pause SBB, and/or block /iclock at the firewall for the window)

# 2. Deploy the code (git pull, or copy the 16 files)

# 3. Inspect, then clear duplicates
php artisan attendance:dedupe --dry      # READ THIS OUTPUT before continuing
php artisan attendance:dedupe

# 4. Apply the schema
php artisan migrate --force

# 5. Rebuild caches
```



```
php artisan optimize:clear
php artisan config:cache
php artisan route:cache

# 6. Restart the queue worker -- NotifyLateArrivals uses the 'mail' queue
php artisan queue:restart

# 7. Verify BEFORE re-enabling the feed
curl -H "X-API-Key: <key>" https://your-server/api/v1/ping
curl "https://your-server/iclock/cdata?SN=<real-serial>&options=all" # old path
still OK

# 8. Re-enable the biometric feed
```

Queue worker

If no worker is running, `QUEUE_CONNECTION=sync` makes the job run inline — notifications still go out, just synchronously. To get the asynchronous behaviour the job exists for, run a worker:

```
php artisan queue:work --queue=mail,default
```

7.3 smartprs-l3 (client build)

This folder is **not a git repository** — there is no `.git` directory, so there is no version history and no git checkout to fall back on. The 16 files were copied in directly. Treat the folder as a build artifact: if it is regenerated from `smartprs2`, the changes come across with it.

7.4 Rollback

What	How
Code — <code>smartprs2</code>	<code>git checkout -- <file></code> for the 4 modified files; delete the 13 new files. Or <code>git stash / git reset --hard</code> if nothing else is in the working tree.
Code — <code>smartprs-l3</code>	No git. Restore from a file backup, or re-copy from <code>smartprs2</code> .
Database	Restore the dump from 7.2 step 0. <code>attendance:dedupe</code> deletes rows and <code>migrate:rollback</code> will NOT bring them back — the dump is the only way back.
Schema only	The migration's <code>down()</code> drops the two unique indexes and the four new columns. Safe, but does not restore deleted duplicates.

8. Git status

Nothing has been committed or pushed

No git command was run during this session — the environment had no shell access to the Windows machine. Every change sits uncommitted in the working tree.

8.1 smartprs2

Item	Value
Repository	Yes
Branch	main
Remote	origin -> https://github.com/ametecsindia/SmartPRS2.git
Last commit	"11-08-2026 update 2" — 11 August 2026, 10:24 UTC
Commits made this session	None
Pushed	No
Covered by .gitignore	None of the changed files — all 17 will appear in git status

Expected git status output

Changes not staged for commit:

```
modified: bootstrap/app.php
modified: routes/web.php
modified: app/Http/Controllers/BiometricConfigController.php
modified: resources/views/layouts/app.blade.php
```

Untracked files:

```
database/migrations/2026_08_16_000100_create_api_keys_table.php
database/migrations/2026_08_16_000200_extend_attendance_logs_for_sbb.php
database/migrations/2026_08_16_000300_create_attendance_pending_table.php
app/Services/ApiKeys.php
app/Services/PunchIngestService.php
app/Http/Middleware/ApiKeyAuth.php
app/Http/Controllers/Api/PublicApiController.php
app/Http/Controllers/ApiKeyController.php
app/Jobs/NotifyLateArrivals.php
routes/api.php
resources/views/settings/api-keys.blade.php
resources/views/settings/pending-punches.blade.php
tests/Feature/SbbIngestTest.php
```

Suggested commit, once the tests pass

```
git checkout -b feature/sbb-ingest
```

```
git add bootstrap/app.php routes/api.php routes/web.php ^
```

```
app/Services/ApiKeys.php app/Services/PunchIngestService.php ^
app/Http/Middleware/ApiKeyAuth.php ^
app/Http/Controllers/Api/PublicApiController.php ^
app/Http/Controllers/ApiKeyController.php ^
app/Http/Controllers/BiometricConfigController.php ^
app/Jobs/NotifyLateArrivals.php ^
database/migrations/2026_08_16_0001*.php ^
database/migrations/2026_08_16_0002*.php ^
database/migrations/2026_08_16_0003*.php ^
resources/views/settings/ resources/views/layouts/app.blade.php ^
tests/Feature/SbbIngestTest.php
```

```
git commit -m "Add authenticated SBB JSON ingest path (/api/v1)"
```

A **branch is recommended over committing straight to `main`**, because the migration performs a one-way destructive dedupe and should be merged deliberately.

8.2 smartprs-l3

Not a git repository. No version control, no history, and no rollback path other than a file backup. Putting the client build under git — or generating it from smartprs2 by script — is recommended so that changes like this are traceable.

9. Open items and recommendations

#	Item	Owner	Priority
1	Run vendor\bin\pest tests/Feature/SbbIngestTest.php. Nothing here is proven until it is green. This is the single most important outstanding action.	Dev	Critical
2	SPA navigation. The two Settings screens are standalone Blade pages on layouts/app at /app/api-keys and /app/pending-punches. The live UI is the SPA in AppController.php (1.1 MB), which was not touched — if that is what users see, the two menu entries need adding there too.	Dev	High
3	Confirm the employee matching strategy. A punch matches by employee_code when SBB sends it, otherwise by the device serial's configured emp_prefix + device_user_id. If SBB will not send employee_code, confirm each device row in Biometric Device Setup has its emp_prefix set.	Ejaz / Dev	High
4	Approve or reject the null-tenant guard described in Section 4.2.	Ejaz	Medium
5	Put smartprs-l3 under version control, or generate it from smartprs2 by script.	Dev	Medium
6	Publish the API contract (Section 3.4) to the SBB team so the Windows service and the server agree on the wire format.	Ejaz	Medium

Prepared 16 August 2026. No database was modified and no commit was made during this session.